Chapter 22

Vector Algebra

Only One Option Correct Type Questions

1. The number of distinct real values of λ, for the vector $-\lambda^2\hat{i}+\hat{j}+\hat{k}$, $\hat{i}-\lambda^2\hat{j}+\hat{k}$ and $\hat{i}+\hat{j}-\lambda^2\hat{k}$ are coplanar, is **[IIT-JEE-2007 (Paper-1)]**

   (A) Zero
   (B) One
   (C) Two
   (D) Three

2. Let $\vec{a}, \vec{b}, \vec{c}$ be unit vectors such that $\vec{a}+\vec{b}+\vec{c}=0$. Which one of the following is correct?

   [IIT-JEE-2007 (Paper-2)]

   (A) $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}=0$
   (B) $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}\neq\vec{0}$
   (C) $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{a}\times\vec{c}\neq\vec{0}$
   (D) $\vec{a}\times\vec{b}$, $\vec{b}\times\vec{c}$, $\vec{c}\times\vec{a}$ are mutually perpendicular

3. The edges of a parallelopiped are of unit length and are parallel to non-coplanar unit vectors $\hat{a}, \hat{b}, \hat{c}$ such that

   $$\hat{a}.\hat{b}=\hat{b}.\hat{c}=\hat{c}.\hat{a}=\frac{1}{2}$$

   Then, the volume of the parallelopiped is **[IIT-JEE-2008 (Paper-1)]**

   (A) $\frac{1}{\sqrt{2}}$
   (B) $\frac{1}{2\sqrt{2}}$
   (C) $\frac{\sqrt{3}}{2}$
   (D) $\frac{1}{\sqrt{3}}$

4. Let two non-collinear unit vectors $\hat{a}$ and $\hat{b}$ form an acute angle. *A* point *P* moves so that at any time *t* the position vector $\overrightarrow{OP}$ (where *O* is the origin) is given by $\hat{a}\cos t+\hat{b}\sin t$. When *P* is farthest from origin *O*, let *M* be the length of $\overrightarrow{OP}$ and $\hat{u}$ be the unit vector along $\overrightarrow{OP}$. Then, **[IIT-JEE-2008 (Paper-2)]**

   (A) $\hat{u}=\frac{\hat{a}+\hat{b}}{|\hat{a}+\hat{b}|}$ and $M=(1+\hat{a}\cdot\hat{b})^{\frac{1}{2}}$
   (B) $\hat{u}=\frac{\hat{a}-\hat{b}}{|\hat{a}-\hat{b}|}$ and $M=(1+\hat{a}\cdot\hat{b})^{\frac{1}{2}}$
   (C) $\hat{u}=\frac{\hat{a}+\hat{b}}{|\hat{a}+\hat{b}|}$ and $M=(1+2\hat{a}\cdot\hat{b})^{\frac{1}{2}}$
   (D) $\hat{u}=\frac{\hat{a}-\hat{b}}{|\hat{a}-\hat{b}|}$ and $M=(1+2\hat{a}\cdot\hat{b})^{\frac{1}{2}}$

5. Let *P*(3, 2, 6) be a point in space and *Q* be a point on the line $\vec{r}=(\hat{i}-\hat{j}+2\hat{k})+\mu(-3\hat{i}+\hat{j}+5\hat{k})$. Then the value of μ for which the vector $\overrightarrow{PQ}$ is parallel to the plane $x - 4y + 3z = 1$ is **[IIT-JEE-2009 (Paper-1)]**

   (A) $\frac{1}{4}$
   (B) $-\frac{1}{4}$
   (C) $\frac{1}{8}$
   (D) $-\frac{1}{8}$

6. If $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are unit vectors such that $(\vec{a} \times \vec{b}).(\vec{c} \times \vec{d}) = 1$ and $\vec{a}.\vec{c} = \frac{1}{2}$, then

[IIT-JEE-2009 (Paper-1)]

(A) $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar
(B) $\vec{b}, \vec{c}, \vec{d}$ are non-coplanar
(C) $\vec{b}, \vec{d}$ are non-parallel
(D) $\vec{a}, \vec{d}$ are parallel and $\vec{b}, \vec{c}$ are parallel

7. Let *P, Q, R* and *S* be the points on the plane with position vectors $-2\hat{i} - \hat{j}$, $4\hat{i}$, $3\hat{i} + 3\hat{j}$ and $-3\hat{i} + 2\hat{j}$ respectively. The quadrilateral *PQRS* must be a **[IIT-JEE-2010 (Paper-1)]**

(A) Parallelogram, which is neither a rhombus nor a rectangle
(B) Square
(C) Rectangle, but not a square
(D) Rhombus, but not a square

8. Two adjacent sides of a parallelogram *ABCD* are given by $\overrightarrow{AB} = 2\hat{i} + 10\hat{j} + 11\hat{k}$ and $\overrightarrow{AD} = -\hat{i} + 2\hat{j} + 2\hat{k}$. The side *AD* is rotated by an acute angle α in the plane of the parallelogram so that *AD* becomes *AD′*. If *AD′* makes a right angle with the side *AB*, then the cosine of the angle α is given by **[IIT-JEE-2010 (Paper-2)]**

(A) $\frac{8}{9}$
(B) $\frac{\sqrt{17}}{9}$
(C) $\frac{1}{9}$
(D) $\frac{4\sqrt{5}}{9}$

9. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = \hat{i} - \hat{j} + \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} - \hat{k}$ be three vectors. A vector $\vec{v}$ in the plane of $\vec{a}$ and $\vec{b}$, whose projection on $\vec{c}$ is $\frac{1}{\sqrt{3}}$, is given by **[IIT-JEE-2011 (Paper-1)]**

(A) $\hat{i} - 3\hat{j} + 3\hat{k}$
(B) $-3\hat{i} - 3\hat{j} - \hat{k}$
(C) $3\hat{i} - \hat{j} + 3\hat{k}$
(D) $\hat{i} + 3\hat{j} - 3\hat{k}$

10. If $\vec{a}$ and $\vec{b}$ are vectors such that $|\vec{a} + \vec{b}| = \sqrt{29}$ and $\vec{a} \times (2\hat{i} + 3\hat{j} + 4\hat{k}) = (2\hat{i} + 3\hat{j} + 4\hat{k}) \times \vec{b}$, then a possible value of $(\vec{a} + \vec{b}).(-7\hat{i} + 2\hat{j} + 3\hat{k})$ is **[IIT-JEE-2012 (Paper-2)]**

(A) 0
(B) 3
(C) 4
(D) 8

11. Let $\overrightarrow{PR} = 3\hat{i} + \hat{j} - 2\hat{k}$ and $\overrightarrow{SQ} = \hat{i} - 3\hat{j} - 4\hat{k}$ determine diagonals of a parallelogram *PQRS* and $\overrightarrow{PT} = \hat{i} + 2\hat{j} + 3\hat{k}$ be another vector. Then the volume of the parallelepiped determined by the vectors $\overrightarrow{PT}, \overrightarrow{PQ}$ and $\overrightarrow{PS}$ is

[JEE (Adv)-2013 (Paper-1)]

(A) 5
(B) 20
(C) 10
(D) 30

12. Let O be the origin and let PQR be an arbitrary triangle. The point S is such that

$$\overrightarrow{OP}\cdot\overrightarrow{OQ}+\overrightarrow{OR}\cdot\overrightarrow{OS}=\overrightarrow{OR}\cdot\overrightarrow{OP}+\overrightarrow{OQ}\cdot\overrightarrow{OS}=\overrightarrow{OQ}\cdot\overrightarrow{OR}+\overrightarrow{OP}\cdot\overrightarrow{OS}$$

Then the triangle PQR has S as its **[JEE (Adv)-2017 (Paper-2)]**

(A) Incentre (B) Circumcentre

(C) Orthocenter (D) Centroid

One or More Option(s) Correct Type Questions

13. The vector(s) which is/are coplanar with vectors $\hat{i}+\hat{j}+2\hat{k}$ and $\hat{i}+2\hat{j}+\hat{k}$, and perpendicular to the vector $\hat{i}+\hat{j}+\hat{k}$ is/are

[IIT-JEE-2011 (Paper-1)]

(A) $\hat{j}-\hat{k}$ (B) $-\hat{j}+\hat{k}$

(C) $\hat{i}-\hat{j}$ (D) $-\hat{j}+\hat{k}$

14. Let $\vec{x}, \vec{y}$ and $\vec{z}$ be three vectors each of magnitude $\sqrt{2}$ and the angle between each pair of them is $\frac{\pi}{3}$. If $\vec{a}$ is a nonzero vector perpendicular to $\vec{x}$ and $\vec{y}\times\vec{z}$ and $\vec{b}$ is a nonzero vector perpendicular to $\vec{y}$ and $\vec{z}\times\vec{x}$, then

[JEE (Adv)-2014 (Paper-1)]

(A) $\vec{b}=(\vec{b}\cdot\vec{z})(\vec{z}-\vec{x})$ (B) $\vec{a}=(\vec{a}\cdot\vec{y})(\vec{y}-\vec{z})$

(C) $\vec{a}\cdot\vec{b}=-(\vec{a}\cdot\vec{y})(\vec{b}\cdot\vec{z})$ (D) $\vec{a}=(\vec{a}\cdot\vec{y})(\vec{z}-\vec{y})$

15. Let ΔPQR be a triangle. Let $\vec{a}=\overrightarrow{QR}$, $\vec{b}=\overrightarrow{RP}$ and $\vec{c}=\overrightarrow{PQ}$. If $|\vec{a}|=12$, $|\vec{b}|=4\sqrt{3}$ and $\vec{b}.\vec{c}=24$, then which of the following is (are) true? **[JEE (Adv)-2015 (Paper-1)]**

(A) $\frac{|\vec{c}|^2}{2}-|\vec{a}|=12$ (B) $\frac{|\vec{c}|^2}{2}+|\vec{a}|=30$

(C) $|\vec{a}\times\vec{b}+\vec{c}\times\vec{a}|=48\sqrt{3}$ (D) $\vec{a}.\vec{b}=-72$

16. Let $\hat{u}=u_1\hat{i}+u_2\hat{j}+u_3\hat{k}$ be a unit vector in $\mathbb{R}^3$ and $\hat{w}=\frac{1}{\sqrt{6}}(\hat{i}+\hat{j}+2\hat{k})$. Given that there exists a vector $\vec{v}$ in $\mathbb{R}^3$ such that $|\hat{u}\times\vec{v}|=1$ and $\hat{w}\cdot(\hat{u}\times\vec{v})=1$. Which of the following statement(s) is(are) correct?

[JEE (Adv)-2016 (Paper-2)]

(A) There is exactly one choice for such $\vec{v}$ (B) There are infinitely many choices for such $\vec{v}$

(C) If $\hat{u}$ lies in the xy-plane, then $|u_1|=|u_2|$ (D) If $\hat{u}$ lies in the xz-plane, then $2|u_1|=|u_3|$

17. Let L_1 and L_2 denote the lines $\vec{r}=\hat{i}+\lambda(-\hat{i}+2\hat{j}+2\hat{k}), \lambda\in\mathbb{R}$ and $\vec{r}=\mu(2\hat{i}-\hat{j}+2\hat{k}), \mu\in\mathbb{R}$ respectively. If L_3 is a line which is perpendicular to both L_1 and L_2 and cuts both of them, then which of the following options describe(s) L_3? **[JEE (Adv)-2019 (Paper-1)]**

(A) $\vec{r}=\frac{1}{3}(2\hat{i}+\hat{k})+t(2\hat{i}+2\hat{j}-\hat{k}), t\in\mathbb{R}$ (B) $\vec{r}=\frac{2}{9}(4\hat{i}+\hat{j}+\hat{k})+t(2\hat{i}+2\hat{j}-\hat{k}), t\in\mathbb{R}$

(C) $\vec{r}=\frac{2}{9}(2\hat{i}-\hat{j}+2\hat{k})+t(2\hat{i}+2\hat{j}-\hat{k}), t\in\mathbb{R}$ (D) $\vec{r}=t(2\hat{i}+2\hat{j}-\hat{k}), t\in\mathbb{R}$

18. Three lines

L_1 : $\vec{r} = \lambda\hat{i},\ \lambda \in \mathbb{R},$

L_2 : $\vec{r} = \hat{k} + \mu\hat{j},\ \mu \in \mathbb{R}$ and

L_3 : $\vec{r} = \hat{i} + \hat{j} + \nu\hat{k},\ \nu \in \mathbb{R}$

are given. For which point(s) Q on L_2 can we find a point P on L_1 and a point R on L_3 so that P, Q and R are collinear? **[JEE (Adv)-2019 (Paper-1)]**

(A) $\hat{k} + \hat{j}$

(B) $\hat{k}$

(C) $\hat{k} + \frac{1}{2}\hat{j}$

(D) $\hat{k} - \frac{1}{2}\hat{j}$

19. Let a and b be positive real numbers. Suppose $\overrightarrow{PQ} = a\hat{i} + b\hat{j}$ and $\overrightarrow{PS} = a\hat{i} - b\hat{j}$ are adjacent sides of a parallelogram $PQRS$. Let $\vec{u}$ and $\vec{v}$ be the projection vectors of $\vec{W} = \hat{i} + \hat{j}$ along $\overrightarrow{PQ}$ and $\overrightarrow{PS}$, respectively. If $|\vec{u}| + |\vec{v}| = |\vec{w}|$ and if the area of the parallelogram $PQRS$ is 8, then which of the following statements is/are TRUE? **[JEE (Adv)-2020 (Paper-2)]**

(A) $a + b = 4$

(B) $a - b = 2$

(C) The length of the diagonal PR of the parallelogram $PQRS$ is 4

(D) $\vec{w}$ is an angle bisector of the vectors $\overrightarrow{PQ}$ and $\overrightarrow{PS}$

20. Let O be the origin and $\overrightarrow{OA} = 2\hat{i} + 2\hat{j} + \hat{k}$, $\overrightarrow{OB} = \hat{i} - 2\hat{j} + 2\hat{k}$ and $\overrightarrow{OC} = \frac{1}{2}(\overrightarrow{OB} - \lambda\overrightarrow{OA})$ for some $\lambda > 0$. If $\left|\overrightarrow{OB} \times \overrightarrow{OC}\right| = \frac{9}{2}$, then which of the following statements is (are) TRUE ? **[JEE (Adv)-2021 (Paper-2)]**

(A) Projection of $\overrightarrow{OC}$ on $\overrightarrow{OA}$ is $-\frac{3}{2}$

(B) Area of the triangle OAB is $\frac{9}{2}$

(C) Area of the triangle ABC is $\frac{9}{2}$

(D) The acute angle between the diagonals of the parallelogram with adjacent sides $\overrightarrow{OA}$ and $\overrightarrow{OC}$ is $\frac{\pi}{3}$

21. Let $\hat{i}, \hat{j}$ and $\hat{k}$ be the unit vectors along the three positive coordinate axes. Let

$\vec{a} = 3\hat{i} + \hat{j} - \hat{k},$
$\vec{b} = \hat{i} + b_2\hat{j} + b_3\hat{k}, \quad b_2, b_3 \in \mathbb{R},$
$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k},\ c_1, c_2, c_3 \in \mathbb{R}$

be three vectors such that $b_2b_3 > 0$, $\vec{a} \cdot \vec{b} = 0$ and

$$\begin{pmatrix} 0 & -c_3 & c_2 \\ c_3 & 0 & -c_1 \\ -c_2 & c_1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} 3-c_1 \\ 1-c_2 \\ -1-c_3 \end{pmatrix}.$$

Then, which of the following is/are TRUE? **[JEE (Adv)-2022 (Paper-2)]**

(A) $\vec{a}\cdot\vec{c}=0$ (B) $\vec{b}\cdot\vec{c}-=0$

(C) $|\vec{b}|>\sqrt{10}$ (D) $|\vec{c}|\le\sqrt{11}$

Linked Comprehension Type Questions

Paragraph for Q.Nos. 22 and 23

Let O be the origin, and $\overrightarrow{OX}$, $\overrightarrow{OY}$, $\overrightarrow{OZ}$ be three unit vectors in the directions of the sides $\overrightarrow{QR}$, $\overrightarrow{RP}$, $\overrightarrow{PQ}$, respectively, of a triangle PQR. **[JEE (Adv)-2017 (Paper-2)]**

Choose the correct answer :

22. $|\overrightarrow{OX}\times\overrightarrow{OY}|=$

(A) sin (P + R) (B) sin 2R

(C) sin (P + Q) (D) sin (Q + R)

23. If the triangle PQR varies, then the minimum value of $\cos(P+Q)+\cos(Q+R)+\cos(R+P)$ is

(A) $-\frac{3}{2}$ (B) $\frac{3}{2}$

(C) $\frac{5}{3}$ (D) $-\frac{5}{3}$

Matrix-Match / Matrix Based Type Questions

24. Match List-I with List-II and select the correct answer using the code given below the lists **[JEE (Adv)-2013 (Paper-2)]**

	List-I		List-II
P.	Volume of parallelepiped determined by vectors $\vec{a},\vec{b}$ and $\vec{c}$ is 2. Then the volume of the parallelepiped determined by vectors $2(\vec{a}\times\vec{b}),3(\vec{b}\times\vec{c})$ and $(\vec{c}\times\vec{a})$ is	1.	100
Q.	Volume of parallelepiped determined by vectors $\vec{a},\vec{b}$ and $\vec{c}$ is 5. Then the volume of the parallelepiped determined by vectors $3(\vec{a}+\vec{b}),(\vec{b}+\vec{c})$ and $2(\vec{c}+\vec{a})$ is	2.	30
R.	Area of a triangle with adjacent sides determined by vectors $\vec{a}$ and $\vec{b}$ is 20. Then the area of the triangle with adjacent sides determined by vectors $(2\vec{a}+3\vec{b})$ and $(\vec{a}-\vec{b})$ is	3.	24

S. Area of a parallelogram with adjacent sides determined by vectors $\vec{a}$ and $\vec{b}$ is 30. Then the area of the parallelogram with adjacent sides determined by vectors $(\vec{a}+\vec{b})$ and $\vec{a}$ is — 4. 60

Codes

	P	Q	R	S
(A)	4	2	3	1
(B)	2	3	1	4
(C)	3	4	1	2
(D)	1	4	3	2

Integer / Numerical Value Type Questions

25. If $\vec{a}$ and $\vec{b}$ are vectors in space given by $\vec{a}=\dfrac{\hat{i}-2\hat{j}}{\sqrt{5}}$ and $\vec{b}=\dfrac{2\hat{i}+\hat{j}+3\hat{k}}{\sqrt{14}}$, then the value of $(2\vec{a}+\vec{b}).[(\vec{a}\times\vec{b})\times(\vec{a}-2\vec{b})]$ is **[IIT-JEE-2010 (Paper-1)]**

26. Let $\vec{a}=-\hat{i}-\hat{k}$, $\vec{b}=-\hat{i}+\hat{j}$ and $\vec{c}=\hat{i}+2\hat{j}+3\hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r}\times\vec{b}=\vec{c}\times\vec{b}$ and $\vec{r}.\vec{a}=0,$ then the value of $\vec{r}.\vec{b}$ is **[IIT-JEE-2011 (Paper-2)]**

27. If $\vec{a}, \vec{b}$ and $\vec{c}$ are unit vectors satisfying $|\vec{a}-\vec{b}|^2+|\vec{b}-\vec{c}|^2+|\vec{c}-\vec{a}|^2=9$, then $|2\vec{a}+5\vec{b}+5\vec{c}|$ is **[IIT-JEE-2012 (Paper-1)]**

28. Consider the set of eight vectors $V=\left\{a\hat{i}+b\hat{j}+c\hat{k} : a, b, c \in \{-1, 1\}\right\}$. Three non-coplanar vectors can be chosen from V in 2^p ways. Then p is **[JEE (Adv)-2013 (Paper-1)]**

29. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-coplanar unit vectors such that the angle between every pair of them is $\dfrac{\pi}{3}$. If $\vec{a}\times\vec{b}+\vec{b}\times\vec{c}=p\vec{a}+q\vec{b}+r\vec{c}$, where p, q and r are scalars, then the value of $\dfrac{p^2+2q^2+r^2}{q^2}$ is **[JEE (Adv)-2014 (Paper-1)]**

30. Let $\vec{a}$ and $\vec{b}$ be two unit vectors such that $\vec{a}\cdot\vec{b}=0$. For some $x, y\in\mathbb{R}$, let $\vec{c}=x\vec{a}+y\vec{b}+\left(\vec{a}\times\vec{b}\right)$. If $|\vec{c}|=2$ and the vector $\vec{c}$ is inclined at the same angle α to both $\vec{a}$ and $\vec{b}$, then the value of $8\cos^2\alpha$ is ______. **[JEE (Adv)-2018 (Paper-1)]**

31. Consider the cube in the first octant with sides OP, OQ and OR of length 1, along the x-axis, y-axis and z-axis, respectively, where $O(0, 0, 0)$ is the origin. Let $S\left(\dfrac{1}{2}, \dfrac{1}{2}, \dfrac{1}{2}\right)$ be the centre of the cube and T be the

vertex of the cube opposite to the origin O such that S lies on the diagonal OT. If $\vec{p}=\overrightarrow{SP}$, $\vec{q}=\overrightarrow{SQ}$, $\vec{r}=\overrightarrow{SR}$ and $\vec{t}=\overrightarrow{ST}$, then the value of $|(\vec{p}\times\vec{q})\times(\vec{r}\times\vec{t})|$ is _____. **[JEE (Adv)-2018 (Paper-2)]**

32. Three lines are given by

$\vec{r}=\lambda\hat{i}, \lambda\in\mathbb{R}$

$\vec{r}=\mu(\hat{i}+\hat{j}), \mu\in\mathbb{R}$ and

$\vec{r}=\nu(\hat{i}+\hat{j}+\hat{k}), \nu\in\mathbb{R}$.

Let the lines cut the plane $x + y + z = 1$ at the points A, B and C respectively. If the area of the triangle ABC is Δ then the value of $(6\Delta)^2$ equals _____ **[JEE (Adv)-2019 (Paper-1)]**

33. Let $\vec{a}=2\hat{i}+\hat{j}-\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}+\hat{k}$ be two vectors. Consider a vector $\vec{c}=\alpha\vec{a}+\beta\vec{b}, \alpha, \beta\in\mathbb{R}$. If the projection of $\vec{c}$ on the vector $\left(\vec{a}+\vec{b}\right)$ is $3\sqrt{2}$, then the minimum value of $\left(\vec{c}-(\vec{a}\times\vec{b})\right).\vec{c}$ equals _____.

[JEE (Adv)-2019 (Paper-2)]

34. In a triangle PQR, let $\vec{a}=\overrightarrow{QR}, \vec{b}=\overrightarrow{RP}$ *and* $\vec{c}=\overrightarrow{PQ}$. If

$$\left|\vec{a}\right|=3,\ \left|\vec{b}\right|=4 \text{ and } \frac{\vec{a}\cdot(\vec{c}-\vec{b})}{\vec{c}\cdot(\vec{a}-\vec{b})}=\frac{\left|\vec{a}\right|}{\left|\vec{a}\right|+\left|\vec{b}\right|}$$

then the value of $\left|\vec{a}\times\vec{b}\right|^2$ is **[JEE (Adv)-2020 (Paper-1)]**

35. Let $\vec{u},\vec{v}$ and $\vec{w}$ be vectors in three-dimensional space, where $\vec{u}$ and $\vec{v}$ are unit vectors which are not perpendicular to each other and

$\vec{u}\cdot\vec{w}=1, \quad \vec{v}\cdot\vec{w}=1, \quad \vec{w}\cdot\vec{w}=4$

If the volume of the parallelopiped, whose adjacent sides are represented by the vectors $\vec{u},\vec{v}$ and $\vec{w}$, is $\sqrt{2}$, then the value of $\left|3\vec{u}+5\vec{v}\right|$ is ________. **[JEE (Adv)-2021 (Paper-1)]**

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